

# ML Specification Report — playground-series-s6e2

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## Heart Disease Prediction · our from-scratch agent (GBDT pool + tree-search v3)

Figures grounded in the competition's `facts.json`, `config.yaml`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

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## Overview

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The task is to rank 270,000 patients by heart-disease probability (metric: ROC-AUC) on a synthetic dataset derived from UCI Heart Disease. We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) refined by a four-stage ladder and a 60-node tree search that ends on a 37-member mega-blend. Our champion scores **CV OOF AUC 0.955529** (submission file rounds to 0.95553); the NVIDIA reproduce-agent, copying a public XGBoost/LightGBM/CatBoost k-fold kernel, reaches **0.9554** — effectively a **tie**, with our agent nudging fractionally ahead.

**Why it matters.** Cardiovascular disease is a leading cause of death worldwide; risk-ranking from routine clinical features supports triage and preventive screening — as decision support, not diagnosis.

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## Data

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**Purpose of Data.** Predict whether a patient has heart disease (Presence vs Absence) — a binary-classification Playground Series episode (S6E2). **Data Format** is clean **tabular CSV**, all numeric columns. **Data Volume** is **630,000 training rows / 270,000 test rows**, with **13 raw predictive features** once `id` and the target are removed.

**Data Quality** is high: **no missing values** in either split, **0 duplicate rows**, and only modest train ↔ test mean shift (largest  $\approx +0.83\%$  on `Number of vessels fluoro`). The class balance is mild — **44.834% positive** (`Heart Disease` = Presence), an **imbalance ratio of 1.23** — which keeps the metric (AUC) clean and

makes class-weighting largely unnecessary (confirmed in training). The strongest EDA signals, all consistent with cardiology intuition, were:

- **Thallium** is the dominant raw predictor (point-biserial  $r \approx +0.606$ ).
- **Chest pain type** ( $r \approx +0.461$ ), **Exercise angina** ( $r \approx +0.442$ ), **Max HR** ( $r \approx -0.441$ ), **Number of vessels fluro** ( $r \approx +0.439$ ), and **ST depression** ( $r \approx +0.431$ ) form the strong mid-tier.
- **Slope of ST** ( $r \approx +0.415$ ) and **Sex** ( $r \approx +0.342$ ) contribute moderately.
- **BP** is essentially uninformative ( $r \approx -0.005$ ); **FBS over 120** ( $r \approx +0.034$ ) and **Cholesterol** ( $r \approx +0.083$ ) are weak — a notable contrast with textbook risk-factor expectations.

**Annotation Guidelines.** The label is `Heart Disease`  $\in \{\text{Absence, Presence}\}$ , mapped to  $\{0, 1\}$  (1 = Presence). Submissions are **positive-class probabilities** scored by AUC — a ranking, not a hard 0/1 decision.

**Feature Set.** Feb's focused **33-feature** set (13 raw UCI columns + 20 engineered) is carried into the July run and used for the XGBoost/CatBoost pool members; the 20 engineered columns were, however, **rejected for the LGB member** by a same-folds prior check (see the Training section), so the champion LGB family runs on the **13 raw features** alone:

| Group              | Features                                                                                                                                                       |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw UCI (13)       | Age, Sex, Chest pain type, BP, Cholesterol, FBS over 120, EKG results, Max HR, Exercise angina, ST depression, Slope of ST, Number of vessels fluro, Thallium  |
| Thresholds / bins  | Age_decade, BP_high, Chol_high, Thallium_7                                                                                                                     |
| Heart-rate derived | HR_reserve, HR_pct_max                                                                                                                                         |
| Interactions       | Age_x_MaxHR, Age_x_STdep, Age_x_Vessels, Chol_x_Age, BP_x_Chol, BP_x_Age, STdep_x_Slope, STdep_x_MaxHR, Vessels_x_Thallium, Angina_x_STdep, ChestPain_x_Angina |
| Composite / ratios | Risk_score, Chol_per_Age, BP_per_Age                                                                                                                           |

**Splitting strategy.** A single canonical **Stratified 5-Fold CV (seed = 42, shuffle)**, shared across all four stages so scores are directly comparable. Because `StratifiedKFold` 's split depends only on the target vector and seed, the July folds reproduce the Feb v1 folds **byte-identically**, letting the reused Feb anchor sit on the same ladder. The leaderboard is closed (late submission returns 403), so all scores here are **CV-only**; Feb's best real submission (v3, on a non-canonical multi-seed scheme) posted Public LB AUC 0.95332 as a loose sanity anchor.

## Models & Architecture

**Purpose of Architecture.** A binary heart-disease classifier that maximizes ROC-AUC. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM as the primary learner, with XGBoost and CatBoost pool members, combined by a weighted blend; the champion is a **37-member mega-blend** discovered at tree-search node #43.

Each member takes the same **Input Format** — a numeric feature matrix (no categorical encoding needed; all columns are already numeric) — of **Input Dimension 13 features** for the champion LGB family (33 for the XGB/CAT members); a 1-D vector with no spatial/sequence structure.

**Architecture Description.** Each GBDT member is an additive ensemble of decision trees grown by histogram-based, leaf-wise gradient boosting (representative stage-2 LGB: ~646 boosting iterations; the Optuna-tuned champion LGB uses `num_leaves` = 102 with a shallow `max_depth` = 4 and strong regularization). The blend layer sits on top as a convex weighted average of members' out-of-fold predictions, with weights found by Dirichlet sampling plus coordinate-ascent refinement. **Model Complexity** is therefore best expressed as trees × leaves rather than a dense parameter count: the champion blends 37 pool members, though the weight search kept only **8 non-zero members** — **all LGB-family variants** of the tuned base-13 LGB, so in practice the mega-blend is an intra-LGB-family variance reduction.

## Training procedures

Training proceeds as a **four-stage ladder** on the shared 5-fold split, each stage measured on the same OOF scale:

| Stage | Configuration                                                                                                             | OOF AUC         |
|-------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1     | Feb v1 LGB+XGB anchor (34 features, fold-identical, reused not retrained)                                                 | 0.95498         |
| 2     | skill pipeline: 33-feature LGB+XGB+CAT blend (weights 0.15 / 0.25 / 0.60)                                                 | 0.955197        |
| 3     | linear iteration: base-13 prior check + <code>is_unbalance</code> ablation + Optuna 40-trial tune + seed-bag, 5-way blend | 0.955510        |
| 4     | tree-search <code>harness_v3</code> (60 nodes, node #43 = 37-member mega-blend)                                           | <b>0.955529</b> |

Two experience-library **priors were re-verified by controlled same-folds comparison** in stage 3: (A) "explicit product/threshold columns add nothing for GBDT" (s3e9) was **confirmed** — the 13 raw features (0.955238) beat the full 33-feature LGB (0.955009) by +0.000229, the *opposite* verdict to s6e1 but the *same* as s5e10; (B) "class-imbalance weighting hurts a ranking metric" (s3e3/s4e1) was **confirmed a third time** — `is_unbalance`=True (0.954993) lost to the default (0.955009) by −0.000016. The **Loss Function** is binary log-loss (LightGBM `binary` objective) with model selection on AUC; the default **unweighted** objective was kept. The **Optimization Algorithm** is gradient boosting (histogram, leaf-wise) — **not** SGD/ADAM — with blend weights optimized by Dirichlet search + coordinate-ascent. The **Learning Rate** is the boosting shrinkage (tuned champion LGB `learning_rate` ≈ 0.0171); there is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

**Training Duration** for the stage-4 tree search was ~**7,901 s of wall-clock over 60 evaluated nodes** (0 failed, 0 dedup rejections; the search stopped on the 60-node hard budget cap); the stage-3 Optuna tune was 40 trials / 515.1 s on a fold-0 proxy. **Training Memory** was not explicitly capped — a 630k × 33 float matrix is comfortable on the single arm64 CPU machine; peak was not separately recorded. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-season experience injection** — distilled

priors from s3e3/s3e9/s5e10/s6e1 carried in and each re-verified on this data. **Data Augmentation** is *N/A* (*tabular*); the analogues are feature engineering and seed-bagging.

**Reproducibility Standards** are strict: fixed seeds (fold seed = 42; Optuna seed and seed-bag seed = 2024 logged), LightGBM determinism flags (`deterministic=True`, `force_row_wise=True`, `fixed_num_threads=16` across all models), pinned from the very first run (the hard-won s6e1 fix). This was a **native v3 run — no replay needed**; the stage-4 OOF-reproduction gate (`06_rebuild_tree_best.py`) retrained all 37 members and matched cached OOFs digit-for-digit ( $\max|\Delta\text{OOF}| = 0$ ), recovering the blend AUC 0.955529 and the stored weights member-for-member.

## Inference procedures

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**Decision Threshold** is *N/A* — AUC is a ranking metric, so we submit raw positive-class probabilities and never threshold (a 0.5 cut would only be needed if hard labels were required). **Inference Duration** was not separately profiled; GBDT scoring of the 270,000 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

## Evaluation & Benchmarking

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**Task for Performance Evaluation.** Rank the test patients by heart-disease probability. **Performance Metrics.** ROC-AUC is the sole competition metric; our stage-4 champion reaches **CV OOF AUC 0.955529** (submission file `0.95553`), climbing  $0.95498 \rightarrow 0.955197 \rightarrow 0.955510 \rightarrow 0.955529$  across the four stages (stage  $1 \rightarrow 4 \approx +0.0575\%$ ). The whole ladder moves within a  $\sim 0.0006$  AUC band — an honest "last-mile" competition where gains are real but small, and the mega-blend still extracted a final  $+0.000019$  over the linear stage's best blend. The Feb v3 prior-season submission (non-canonical multi-seed CV) posted Public LB 0.95332 — reference only, as this run is CV-only.

**Performance Benchmarking.** The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel:

| Agent     | Approach                                                                                     | CV AUC         | Note         |
|-----------|----------------------------------------------------------------------------------------------|----------------|--------------|
| NVIDIA    | reproduces <code>kospintr/heart-xgb-lightgbm-catb-baseline-k-fold</code> (verbatim-ish, CPU) | 0.9554         | tie          |
| Our agent | from-scratch GBDT pool + tree-search v3                                                      | <b>0.95553</b> | nudges ahead |

The result is effectively a **dead heat**, with our from-scratch agent fractionally ahead of the reproduced public kernel — consistent with the newer-season (S5–S6) pattern where the reproduce-vs-originate gap collapses, in contrast to the small public-kernel edge seen on older S3-era problems. Both numbers are leak-free.

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## Final-Architecture Re-Run (v5 pipeline, Stage 0.5 in-lane) — 2026-08

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All 20 baseline competitions were re-run in an isolated root ( `~/benchruns/myagent-rerun` ) under one frozen architecture ( `docs/rerun_manifest.json` ): every lane runs the **full v5 pipeline including the Stage 0.5 problem dossier** ( `dossier.json` written in-lane before modeling ), holds no credentials, and is checked by a per-lane transcript audit. Leaderboard scores for the re-run were produced by the separate credentialed operator scoring step and are recorded in `benchmark_results/rerun_three_way.csv` ; the figures below revise nothing in the sections above.

- **Lane**: ran and was scored.
- re-run local CV (lane `STATUS.md` ): **OOF ROC-AUC 0.955418** — 27-member blend (6 with non-zero weight), tree-search node #43, 5-fold StratifiedKFold (shuffle, seed 42).
- Public / private leaderboard for this re-run ( `benchmark_results/rerun_three_way.csv` ): **Public 0.95354 / Private 0.95505**; `winner_priv` = **aide** (AIDE 0.95516, NVIDIA 0.95508).